Mathematical Formulation of the Constant-Strength Vortex Panel Method

This document derives the closed-form influence coefficients K_{ij} and L_{ij} used by the constant-strength vortex panel solver, in the sign conventions of the code. The potential-flow formulation follows Anderson [1]; thin airfoil theory, used as a reference in the technical report, is taken from the same text and is not re-derived here. Verification and results are reported separately in the technical report. Original handwritten notes are included in the project repository for viewing.

Panel Geometry and Conventions

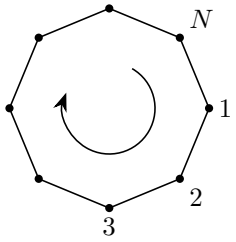
The airfoil surface is discretized into N straight panels, ordered in the clockwise direction (Figure 1a). Each panel a is defined by two *boundary points*, the panel nodes (X_a, Y_a) and (X_{a+1}, Y_{a+1}). Its *control point* (x_a, y_a) is the panel midpoint, where the boundary condition is enforced. With $\Delta X = X_{a+1} - X_a$ and $\Delta Y = Y_{a+1} - Y_a$, the panel angle and length are

$$\varphi_a = \operatorname{atan2}(\Delta Y, \Delta X), \qquad S_a = \|(\Delta X, \Delta Y)\|_2 .$$

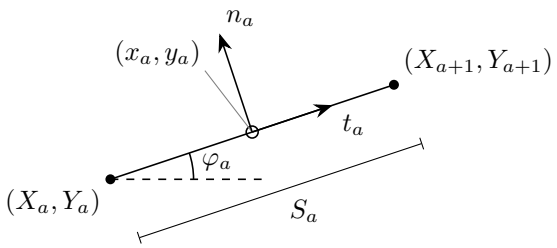
The normal direction is rotated 90° from the panel, and β_a is the angle between the freestream and the normal:

$$\delta_a = \varphi_a + 90^\circ, \qquad \beta_a = \delta_a - \alpha \quad (\beta_a = \delta_a \text{ for } \alpha = 0).$$

With clockwise ordering, δ_a points out of the body. For example, on the lower surface the panels run from the trailing edge toward the leading edge ($\varphi_a \approx 180^\circ$), so $\delta_a \approx 270^\circ$ points downward, away from the airfoil.



(a) Clockwise panel ordering



(b) Geometry of panel a

Figure 1: Panel discretization. Boundary points are the panel nodes; the control point is the panel midpoint, with normal n_a at angle $\delta_a = \varphi_a + 90^\circ$ and tangent t_a at angle φ_a. The panel length is S_a.

Velocity Potential

Consider a point P in the flow field. Recall that the velocity potential of a point vortex of strength Γ is

$$\phi = -\frac{\Gamma}{2\pi}\theta,$$

where θ is the polar angle measured from the vortex to P. For N point vortices $\Gamma_1, \ldots, \Gamma_N$, flow superposition gives

$$\phi_P = \phi_1 + \phi_2 + \cdots + \phi_N = -\sum_{i=1}^{N} \frac{\Gamma_i}{2\pi}\,\theta_{iP}.$$

Letting the number of vortices become infinite, the discrete strengths become a continuous contour with local vortex strength $\gamma(s)$ per unit length along a curve from a to b:

$$\phi_P = -\int_a^b \frac{\gamma(s)}{2\pi}\,\theta\,ds.$$

When we break the continuous line into "panels", we make the assumption that each panel has a constant vortex strength. Hence the j-th panel has vortex strength γ_j. Placing a vortex sheet on each panel j with *constant* strength γ_j allows the strength to come outside the integral. Adding the uniform freestream at angle of attack α,

$$\phi_P = V_\infty \cos(\alpha)\, x + V_\infty \sin(\alpha)\, y - \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \theta_{Pj}\, ds_j. \tag{1}$$

Note, we are still integrating the polar angle across each panel from point P.

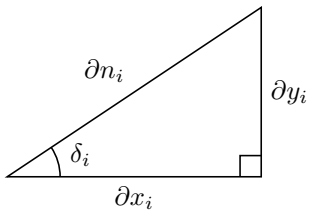
Normal Velocity and the Coefficient K_{ij}

Evaluating (1) at the control point of an arbitrary panel i,

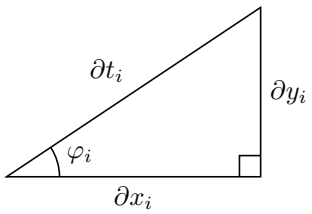
$$\phi_i = V_\infty \cos(\alpha)\, x_i + V_\infty \sin(\alpha)\, y_i - \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \theta_{ij}\, ds_j.$$

The normal velocity at the ith panel is the derivative with respect to n_i:

$$V_{n,i} = \frac{\partial \phi_i}{\partial n_i} = V_\infty \cos(\alpha) \frac{\partial x_i}{\partial n_i} + V_\infty \sin(\alpha) \frac{\partial y_i}{\partial n_i} - \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i}\, ds_j.$$



(a) Normal direction



(b) Tangential direction

Figure 2: Components of the normal and tangential directions at control point i: $\partial x_i/\partial n_i = \cos\delta_i$, $\partial y_i/\partial n_i = \sin\delta_i$; $\partial x_i/\partial t_i = \cos\varphi_i$, $\partial y_i/\partial t_i = \sin\varphi_i$.

From Figure 2a, $\cos\delta_i = \partial x_i/\partial n_i$ and $\sin\delta_i = \partial y_i/\partial n_i$, so the freestream terms combine:

$$V_{n,i,\infty} = V_\infty \left(\cos\alpha\cos\delta_i + \sin\alpha\sin\delta_i\right) = V_\infty \cos(\delta_i - \alpha) = V_\infty \cos\beta_i$$

$$\Rightarrow \quad V_{n,i} = V_\infty \cos(\beta_i) - \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i}\, ds_j. \tag{2}$$

We want an expression for

$$K_{ij} = \int_j \frac{\partial \theta_{ij}}{\partial n_i}\, ds_j.$$

The angle θ_{ij} is measured from a point (x_j, y_j) on panel j to the control point (x_i, y_i) of panel i:

$$\theta_{ij} = \tan^{-1}\left(\frac{y_i - y_j}{x_i - x_j}\right) \quad \Rightarrow \quad K_{ij} = \int_j \frac{\partial}{\partial n_i} \left[\tan^{-1}\left(\frac{y_i - y_j}{x_i - x_j}\right)\right] ds_j.$$

By the chain rule,

$$\frac{\partial}{\partial n_i}\tan^{-1}\left(\frac{y_i-y_j}{x_i-x_j}\right)=\left[\frac{1}{1+\left(\frac{y_i-y_j}{x_i-x_j}\right)^2}\right]\left[\frac{(x_i-x_j)\left(\frac{\partial y_i}{\partial n_i}-\frac{\partial y_j}{\partial n_i}\right)-(y_i-y_j)\left(\frac{\partial x_i}{\partial n_i}-\frac{\partial x_j}{\partial n_i}\right)}{(x_i-x_j)^2}\right],$$

where

$$\frac{1}{1+\left(\frac{y_i-y_j}{x_i-x_j}\right)^2}=\frac{(x_i-x_j)^2}{(x_i-x_j)^2+(y_i-y_j)^2}.$$

The points of panel j do not move with the control point of panel i, so $\partial x_j/\partial n_i=\partial y_j/\partial n_i=0$, and

$$\frac{\partial\theta_{ij}}{\partial n_i}=\frac{(x_i-x_j)\frac{\partial y_i}{\partial n_i}-(y_i-y_j)\frac{\partial x_i}{\partial n_i}}{(x_i-x_j)^2+(y_i-y_j)^2}.$$

Substituting $\cos\delta_i=\partial x_i/\partial n_i$ and $\sin\delta_i=\partial y_i/\partial n_i$,

$$K_{ij}=\int_j\frac{(x_i-x_j)\sin\delta_i-(y_i-y_j)\cos\delta_i}{(x_i-x_j)^2+(y_i-y_j)^2}\,ds_j. \tag{3}$$

Analytic Evaluation of K_{ij}

Substitute the coordinates of panel j in terms of the panel arc length s_j and its boundary point (X_j,Y_j). Note that $s_j\in[0,S_j]$ where S_j is the length of the panel:

$$x_j=X_j+s_j\cos\varphi_j,\qquad y_j=Y_j+s_j\sin\varphi_j,\qquad 0\le s_j\le S_j.$$

Also note that, since $\delta=\varphi+90^\circ$,

$$\cos\delta=\cos(\varphi+90^\circ)=-\sin\varphi,\qquad \sin\delta=\sin(\varphi+90^\circ)=\cos\varphi.$$

Let $K_{ij}=\int_j\frac{f(s_j)}{g(s_j)}\,ds_j$, with $f(s_j)$ the numerator and $g(s_j)$ the denominator of (3).

Numerator.

$$\begin{aligned}f(s_j)&=\big(x_i-(X_j+s_j\cos\varphi_j)\big)\cos\varphi_i+\big(y_i-(Y_j+s_j\sin\varphi_j)\big)\sin\varphi_i\\&=x_i\cos\varphi_i-X_j\cos\varphi_i-s_j\cos\varphi_i\cos\varphi_j+y_i\sin\varphi_i-Y_j\sin\varphi_i-s_j\sin\varphi_i\sin\varphi_j\\&=(x_i-X_j)\cos\varphi_i+(y_i-Y_j)\sin\varphi_i-s_j\left(\cos\varphi_i\cos\varphi_j+\sin\varphi_i\sin\varphi_j\right)\\&=(x_i-X_j)\cos\varphi_i+(y_i-Y_j)\sin\varphi_i-s_j\cos(\varphi_i-\varphi_j).\end{aligned}$$

The numerator is therefore linear in s_j, $f=Cs_j+D$, with

$$C=-\cos(\varphi_i-\varphi_j),\qquad D=(x_i-X_j)\cos\varphi_i+(y_i-Y_j)\sin\varphi_i.$$

Denominator.

$$g(s_j)=\big(x_i-(X_j+s_j\cos\varphi_j)\big)^2+\big(y_i-(Y_j+s_j\sin\varphi_j)\big)^2.$$

Using $(a-b-c)^2=a^2+b^2+c^2-2ab-2ac+2bc$,

$$\begin{aligned}g(s_j)&=x_i^2+X_j^2+s_j^2\cos^2\varphi_j-2x_iX_j-2s_jx_i\cos\varphi_j+2X_js_j\cos\varphi_j\\&\quad+y_i^2+Y_j^2+s_j^2\sin^2\varphi_j-2y_iY_j-2s_jy_i\sin\varphi_j+2Y_js_j\sin\varphi_j\\&=s_j^2+2s_j\big[(-x_i+X_j)\cos\varphi_j+(-y_i+Y_j)\sin\varphi_j\big]+(x_i-X_j)^2+(y_i-Y_j)^2\\&=s_j^2+2As_j+B,\end{aligned}$$

where

$$A = -(x_i - X_j)\cos\varphi_j - (y_i - Y_j)\sin\varphi_j, \qquad B = (x_i - X_j)^2 + (y_i - Y_j)^2.$$

Integration. Rewriting the numerator as $C(s_j + A) + (D - AC)$,

$$\begin{aligned} K_{ij} &= \int_0^{S_j} \frac{Cs_j + D}{s_j^2 + 2As_j + B}\, ds_j = \int_0^{S_j} \frac{C(s_j + A) + (D - AC)}{s_j^2 + 2As_j + B}\, ds_j \\ &= \frac{C}{2}\int_0^{S_j} \frac{2s_j + 2A}{s_j^2 + 2As_j + B}\, ds_j + \int_0^{S_j} \frac{D - AC}{s_j^2 + 2As_j + B}\, ds_j. \end{aligned}$$

Let $u = s_j^2 + 2As_j + B$, so $du = (2s_j + 2A)\, ds_j$, and note that completing the square gives

$$s_j^2 + 2As_j + B = s_j^2 + 2As_j + A^2 + B - A^2 = (s_j + A)^2 + (B - A^2).$$

First term:

$$K_{ij,1} = \frac{C}{2}\int \frac{1}{u}\, du = \frac{C}{2}\ln(u)\Big|_{s_j=0}^{s_j=S_j} = \frac{C}{2}\ln\big(S_j^2 + 2AS_j + B\big) - \frac{C}{2}\ln(B) = \frac{C}{2}\ln\left(\frac{S_j^2 + 2AS_j + B}{B}\right).$$

Second term: let $E^2 = B - A^2$. Then

$$\begin{aligned} K_{ij,2} &= \int_0^{S_j} \frac{D - AC}{(s_j + A)^2 + (B - A^2)}\, ds_j = (D - AC)\int_0^{S_j} \frac{ds_j}{(s_j + A)^2 + E^2} \\ &= \frac{D - AC}{E}\tan^{-1}\left(\frac{s_j + A}{E}\right)\Big|_0^{S_j} = \frac{D - AC}{E}\left[\tan^{-1}\left(\frac{S_j + A}{E}\right) - \tan^{-1}\left(\frac{A}{E}\right)\right]. \end{aligned}$$

And therefore, putting the two terms together, we have

$$\boxed{K_{ij} = \frac{C}{2}\ln\left(\frac{S_j^2 + 2AS_j + B}{B}\right) + \frac{D - AC}{E}\left[\tan^{-1}\left(\frac{S_j + A}{E}\right) - \tan^{-1}\left(\frac{A}{E}\right)\right]} \tag{4}$$

$$\begin{aligned} &A = -(x_i - X_j)\cos\varphi_j - (y_i - Y_j)\sin\varphi_j, \quad && B = (x_i - X_j)^2 + (y_i - Y_j)^2, && E = \sqrt{B - A^2}, \\ &C = -\cos(\varphi_i - \varphi_j), && D = (x_i - X_j)\cos\varphi_i + (y_i - Y_j)\sin\varphi_i. \end{aligned}$$

For $i = j$, $C = -\cos(0) = -1$. Using the midpoint relations $x_i - X_i = \frac{S_i}{2}\cos\varphi_i$ and $y_i - Y_i = \frac{S_i}{2}\sin\varphi_i$ (derived for L_{ii} below), $D = \frac{S_i}{2}$, $A = -\frac{S_i}{2}$ and $B = \frac{S_i^2}{4}$, so the denominator is $(s_j - S_i/2)^2$ and the integrand reduces to

$$\frac{-s_j + S_i/2}{(s_j - S_i/2)^2} = -\frac{1}{s_j - S_i/2}.$$

This is odd about the panel midpoint, so the two halves of the panel cancel and $K_{ii} = 0$ as a Cauchy principal value. Note that $E = \sqrt{B - A^2}$ is zero whenever control point i lies on the line through panel j (always for $i = j$), so (4) is evaluated only for $E > 0$; K_{ii} is set directly.

Tangential Velocity and the Coefficient L_{ij}

Differentiating (1) with respect to the tangent direction t_i instead,

$$\bar{V}_{t,i} = \frac{\partial\phi_i}{\partial t_i} = V_\infty\sin(\beta_i) - \sum_{j\neq i}^{N}\frac{\gamma_j}{2\pi}L_{ij}, \qquad L_{ij} = \int_j \frac{\partial\theta_{ij}}{\partial t_i}\, ds_j,$$

where, from Figure 2b, $\partial x_i/\partial t_i = \cos\varphi_i$ and $\partial y_i/\partial t_i = \sin\varphi_i$. The freestream term follows from

$$V_\infty(\cos\alpha\cos\varphi_i + \sin\alpha\sin\varphi_i) = V_\infty\cos(\varphi_i - \alpha) = V_\infty\sin(\beta_i), \qquad \text{since } \beta_i = \varphi_i + 90^\circ - \alpha.$$

The same steps as for K_{ij} give a numerator $(x_i - x_j)\sin\varphi_i - (y_i - y_j)\cos\varphi_i$, which is again linear in s_j. The denominator, and therefore A, B and E, are unchanged. Similarly,

$$\boxed{L_{ij} = \frac{C}{2}\ln\left(\frac{S_j^2 + 2AS_j + B}{B}\right) + \frac{D - AC}{E}\left[\tan^{-1}\left(\frac{S_j + A}{E}\right) - \tan^{-1}\left(\frac{A}{E}\right)\right]} \tag{5}$$

with A, B, E as before and

$$C = \sin(\varphi_j - \varphi_i), \qquad D = (x_i - X_j)\sin\varphi_i - (y_i - Y_j)\cos\varphi_i.$$

Note that when $i = j$, $C = \sin(\varphi_i - \varphi_i) = 0$. Using the fact that (x_i, y_i) is the midpoint between (X_i, Y_i) and (X_{i+1}, Y_{i+1}), we can write:

$$(x_i - X_i) = \frac{(X_{i+1} - X_i)}{2}$$

Since $\Delta X, \Delta Y, S_i$ form a right triangle, we can use the fact that

$$\cos(\varphi_i) = \frac{(X_{i+1} - X_i)}{S_i}, \quad \sin(\varphi_i) = \frac{(Y_{i+1} - Y_i)}{S_i}$$

Then, we have that

$$(x_i - X_i) = \frac{(X_{i+1} - X_i)}{2} = \frac{S_i}{2}\cdot\frac{(X_{i+1} - X_i)}{S_i} = \frac{S_i}{2}\cos(\varphi_i), \quad \text{likewise } \ (y_i - Y_i) = \frac{S_i}{2}\sin(\varphi_i)$$

By substitution,

$$D = \frac{S_i}{2}\cos\varphi_i\sin\varphi_i - \frac{S_i}{2}\sin\varphi_i\cos\varphi_i = 0$$

With $C = D = 0$, the numerator $Cs_j + D$ of the integrand vanishes identically. The denominator is $(s_j - S_i/2)^2$, as for K_{ii}, so the integrand is zero everywhere except at the single point $s_j = S_i/2$. Therefore, the diagonal terms of L are 0. As with K_{ii}, (5) cannot be used directly here because $E = 0$ (its second term becomes $0/0$), so L_{ii} is set directly.

Now, consider the fact that $\gamma = u_1 - u_2$ where u_1, u_2 are the tangential velocities immediately outside and inside the vortex sheet [1, Eq. 4.8]. Now, write

$$2u_1 = u_1 + u_2 + u_1 - u_2 \implies u_1 = \frac{u_1 + u_2}{2} + \frac{u_1 - u_2}{2} = \bar{u} + \frac{\gamma}{2}$$

Here u_1 is the tangential velocity just outside the panel, $u_1 = V_{t,i}$. To identify $\bar{u}$, note that the freestream and the panels $j \neq i$ are smooth at panel i, so they contribute equally just outside and just inside it. Panel i's own contribution is antisymmetric across the panel (its induced tangential velocity is odd in the normal distance from the panel), so its outside and inside values average to zero, which is also its value on the panel itself, since $L_{ii} = 0$. The on-panel evaluation $\bar{V}_{t,i}$ therefore gives the average:

$$\bar{u} = \bar{V}_{t,i} = V_\infty\sin(\beta_i) - \sum_{j\neq i}^{N}\frac{\gamma_j}{2\pi}L_{ij}$$

$$\implies V_{t,i} = V_\infty\sin(\beta_i) + \frac{\gamma_i}{2} - \sum_{j\neq i}^{N}\frac{\gamma_j}{2\pi}L_{ij} \tag{6}$$

And it follows that the coefficient of pressure C_p of the i-th panel is given by:

$$C_{p,i} = 1 - \left(\frac{V_{t,i}}{V_\infty}\right)^2 \tag{7}$$

Linear System

Using the zero-normal-velocity (flow tangency) condition at every control point, (2) becomes

$$V_{n,i} = V_\infty \cos(\beta_i) - \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} K_{ij} = 0 \quad \Rightarrow \quad \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} K_{ij} = V_\infty \cos(\beta_i), \qquad i = 1, \ldots, N.$$

In matrix form, with $K_{ii} = 0$,

$$\frac{1}{2\pi} \begin{pmatrix} 0 & K_{12} & K_{13} & \cdots & K_{1N} \\ K_{21} & 0 & K_{23} & \cdots & K_{2N} \\ \vdots & & \ddots & & \vdots \\ K_{N1} & K_{N2} & K_{N3} & \cdots & 0 \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_N \end{pmatrix} = \begin{pmatrix} V_\infty \cos\beta_1 \\ V_\infty \cos\beta_2 \\ \vdots \\ V_\infty \cos\beta_N \end{pmatrix}.$$

When we apply a singular value decomposition on the K matrix, it turns out to be exactly singular. Hence, there is no unique solution, so we must enforce the *Kutta condition* onto the matrix to enforce a physical solution. The Kutta condition states that the vortex strengths of the upper panel adjacent to the TE and the lower panel adjacent to the TE sum to 0:

$$\gamma_1 + \gamma_N = 0$$

Since the repository implementation includes a blunt panel, we actually use the upper and lower panels adjacent to the blunt panel. Assuming the blunt panel is the N-th panel, we modify the Kutta condition accordingly:

$$\gamma_1 + \gamma_{N-1} = 0$$

To preserve the $N \times N$ structure of the matrix, we need to replace an existing linear equation. If the discretization used a closed trailing edge, we might replace the flow-tangency equation of one panel on the upper or lower surface, somewhere near the middle of the airfoil. Since the blunt TE geometry presents an equation for the blunt panel, we may replace it while still applying the Kutta condition on the adjacent upper and lower panels. Our modified matrix equation is:

$$\frac{1}{2\pi} \begin{pmatrix} 0 & K_{12} & \cdots & K_{1,N-1} & K_{1N} \\ K_{21} & 0 & \cdots & K_{2,N-1} & K_{2N} \\ \vdots & & \ddots & \vdots & \vdots \\ K_{N-1,1} & K_{N-1,2} & \cdots & 0 & K_{N-1,N} \\ 1 & 0 & \cdots & 1 & 0 \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_{N-1} \\ \gamma_N \end{pmatrix} = \begin{pmatrix} V_\infty \cos\beta_1 \\ V_\infty \cos\beta_2 \\ \vdots \\ V_\infty \cos\beta_{N-1} \\ 0 \end{pmatrix}.$$

Lift

The total circulation is the sum of the panel strengths times their lengths, $\Gamma = \sum_{j=1}^{N} \gamma_j S_j$. From the Kutta–Joukowski theorem,

$$L' = \rho_\infty V_\infty \Gamma \quad \Rightarrow \quad c_l \, q_\infty c = \rho_\infty V_\infty \Gamma \quad \Rightarrow \quad c_l = \frac{2\Gamma}{V_\infty c},$$

which reduces to $c_l = 2\Gamma$ for the normalized case $V_\infty = 1$, $c = 1$.

The lift coefficient is also computed by summing the pressure across the airfoil panels. From the coefficient of pressure (7) and the known outward normal angle (δ), we can calculate the coefficients of normal and axial forces:

$$c_n = -\frac{1}{c} \sum_{i=1}^{N} C_{p,i} \cdot S_i \cdot \sin(\delta_i)$$

$$c_a = -\frac{1}{c} \sum_{i=1}^{N} C_{p,i} \cdot S_i \cdot \cos(\delta_i)$$

And then we can project these coefficients into the coefficient of lift with angle of attack effects:

$$c_l(\alpha) = c_n \cos(\alpha) - c_a \sin(\alpha)$$

The same projection along the freestream gives the pressure drag, $c_d = c_n \sin(\alpha) + c_a \cos(\alpha)$, which should be close to zero in inviscid flow (d'Alembert's paradox). Its departure from zero is a check on the discretization.

References

[1] J. D. Anderson, Jr. and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. New York, NY: McGraw-Hill, 2024.